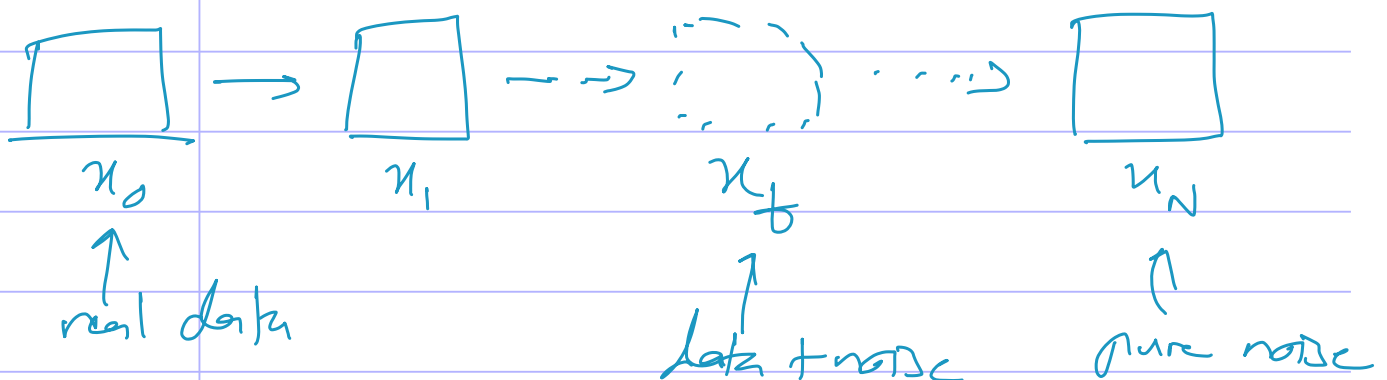


Diffusion Transformer (DiT)

Key Contributions

- replace U-net backbone in diffusion models w/ transformers
- show that diffusion quality increases w/ scale (compute, model size, token size)

Refresher on Diffusion Formulation



$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\alpha_t} x_0, (1 - \alpha_t) \mathbf{I})$$

→ ①

* This is a random sampling which is not differentiable. So cannot train w backprop.

* Reparameterization trick
→ move randomness to an external variable that doesn't depend on model parameters.

* to sample from $n \sim \mathcal{N}(\mu, \sigma^2)$
we can sample standard noise from $\mathcal{N}(0, 1)$ and scale it by σ and add μ .

$$n = \mu + \sigma \epsilon$$

* for over $eq^{\circ} \textcircled{1}$,

$$n_t = \sqrt{\alpha_t} n_0 + \sqrt{1 - \alpha_t} \epsilon_t$$

where $\epsilon_t \sim \mathcal{N}(0, I)$

* Diffusion models are trained to invert the noise process

$$p_{\theta}(u_{t-1} | u_t) = \mathcal{N}(\mu_{\theta}(u_t), \Sigma_{\theta}(u_t))$$

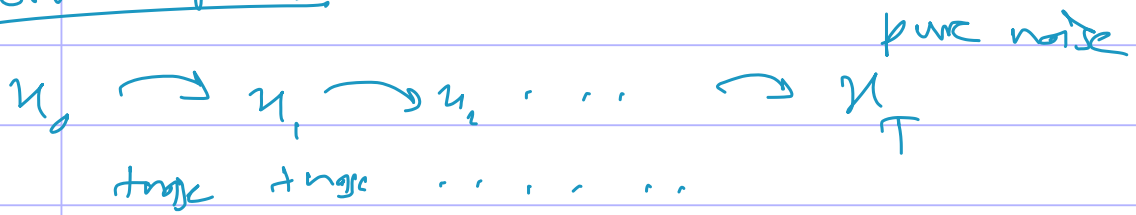
NN are trained to predict the statistics of p_{θ}

* reverse process model is trained to variational lower bound the log likelihood of u_0

$$\mathcal{L}(\theta) = -\log p(u_0 | u_1)$$

$$\neq \sum_t \mathbb{D}_{KL} \left[q^*(u_{t-1} | u_t, u_0) \parallel p_{\theta}(u_{t-1} | u_t) \right]$$

forward process



* goal of model p_θ is to learn reverse process

$$p_\theta(z_{t-1} | z_t) = \mathcal{N}(\mu_\theta(z_t), \Sigma_\theta(z_t))$$

* instead of predicting z_{t-1} directly, NN predicts statistics μ & Σ of distribution of z_{t-1} given z_t

Variational Lower Bound (VLB) aka ELBO

* To train the model, we want to maximize the likelihood of the real data $p(z_0)$.

* Since calculating $p(z_0)$ directly is intractable, we optimize VLB

$$* \mathcal{L}(\theta) = -p(x_0 | u_1) + \sum_t \mathbb{D}_{KL} \left[q^*(x_{t-1} | u_t, u) \parallel p_\theta(x_{t-1} | u_t) \right]$$

* $q(x_{t-1} | u_t, u_0) \Rightarrow$ forward posterior

- represents the distribution of x_{t-1} if we already know both current noisy state x_t and original clean image x_0

- b/c fwd process is Gaussian, this fwd posterior is also Gaussian $\Rightarrow$ a known closed-form mean & variance

* $p_\theta(x_{t-1} | u_t) \Rightarrow$ what the model predicts.

Goal: by minimizing KL divergence we are forcing the model's predicted distribution p_θ to match ground-truth distribution q

* In original DDPM, covariance of noise Σ_θ is fixed constant.

* here we learn Σ_θ as it improves results

* we 2 losses

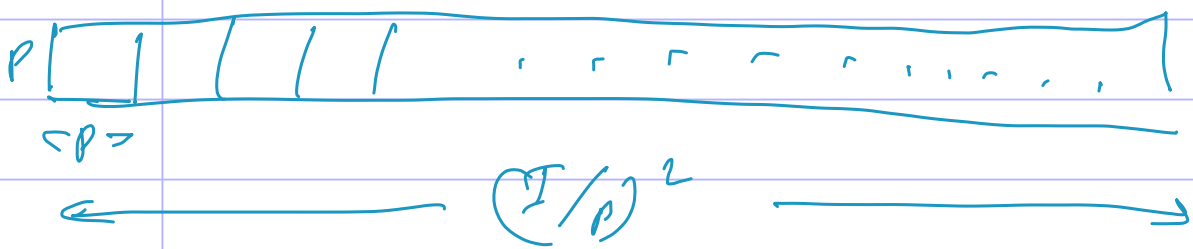
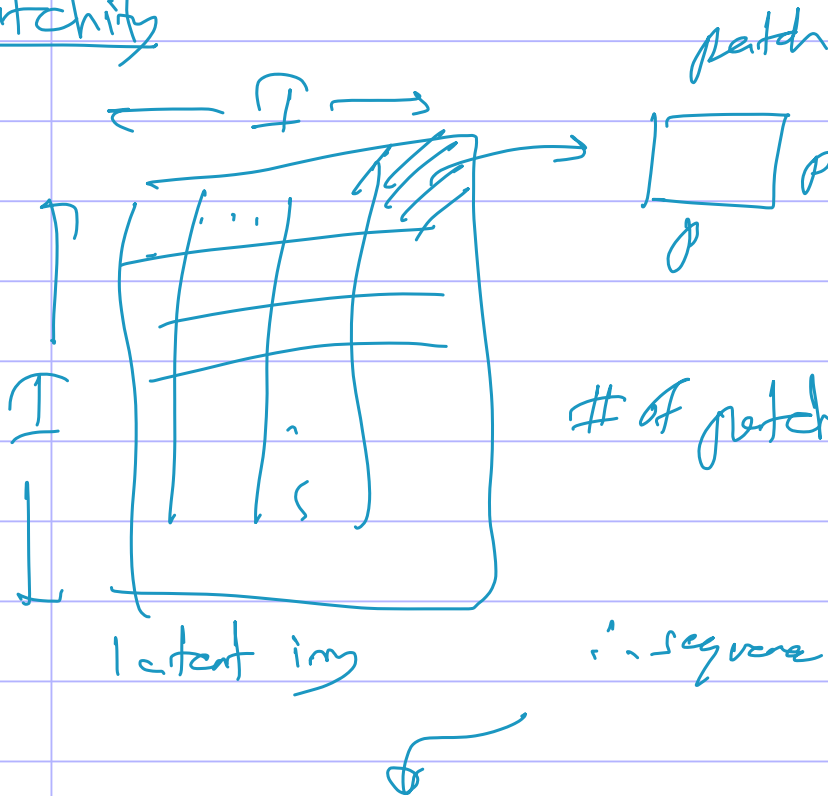
$$\mathcal{L}_{\text{simple}}(\theta) = \|\Sigma_\theta(x_t) - \epsilon_t\|_2^2$$

to train μ_θ

full KL div loss to train Σ_θ

29/3/2026

Patching



Pixel space

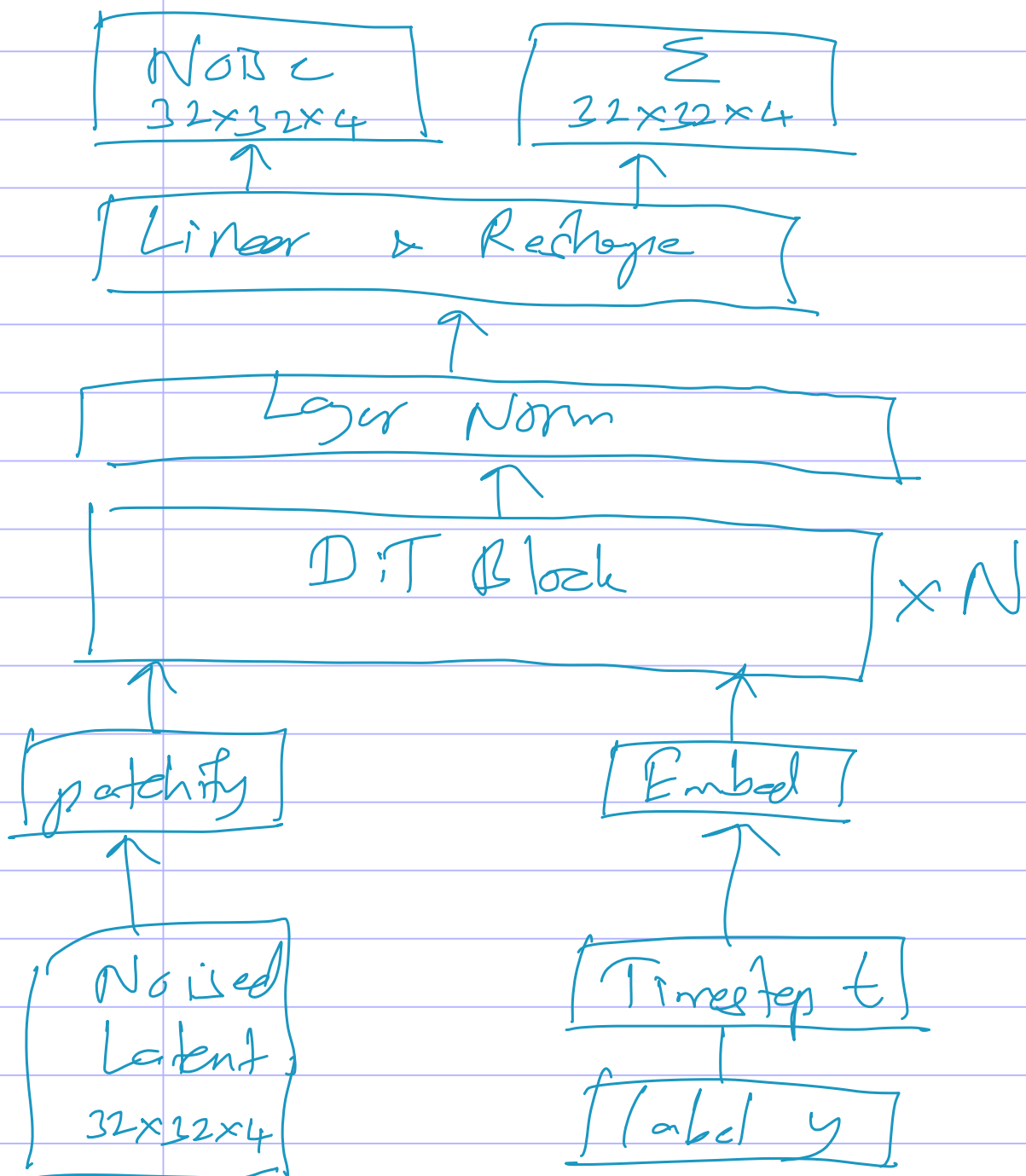
* $\wedge$ img size = $256 \times 256 \times 3$

* latent space size = $(32 \times 32 \times 4)$

patching converts this latent space tensor into patches as described above

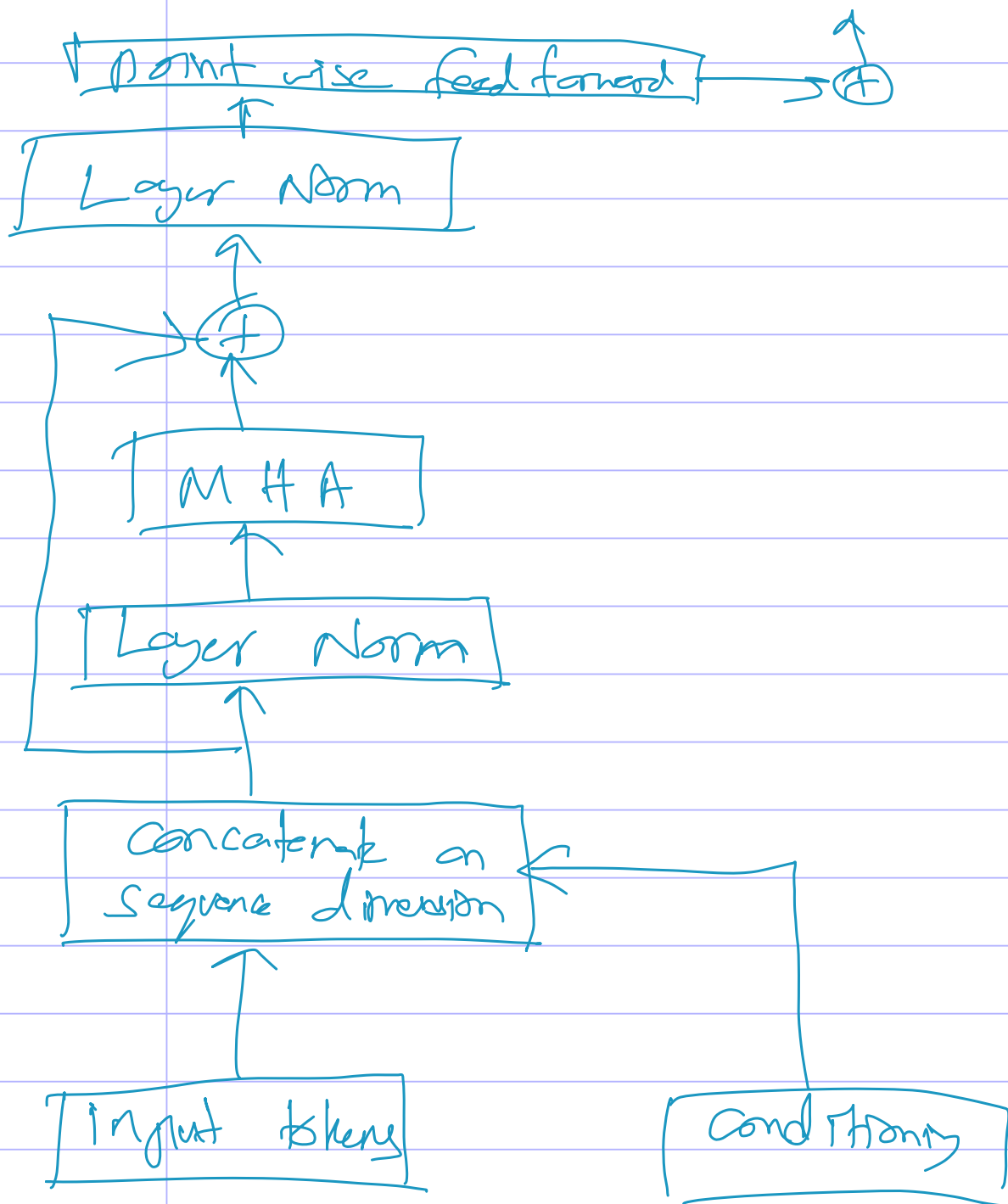
* after patching add size-aware positional embedding

DiT Architecture

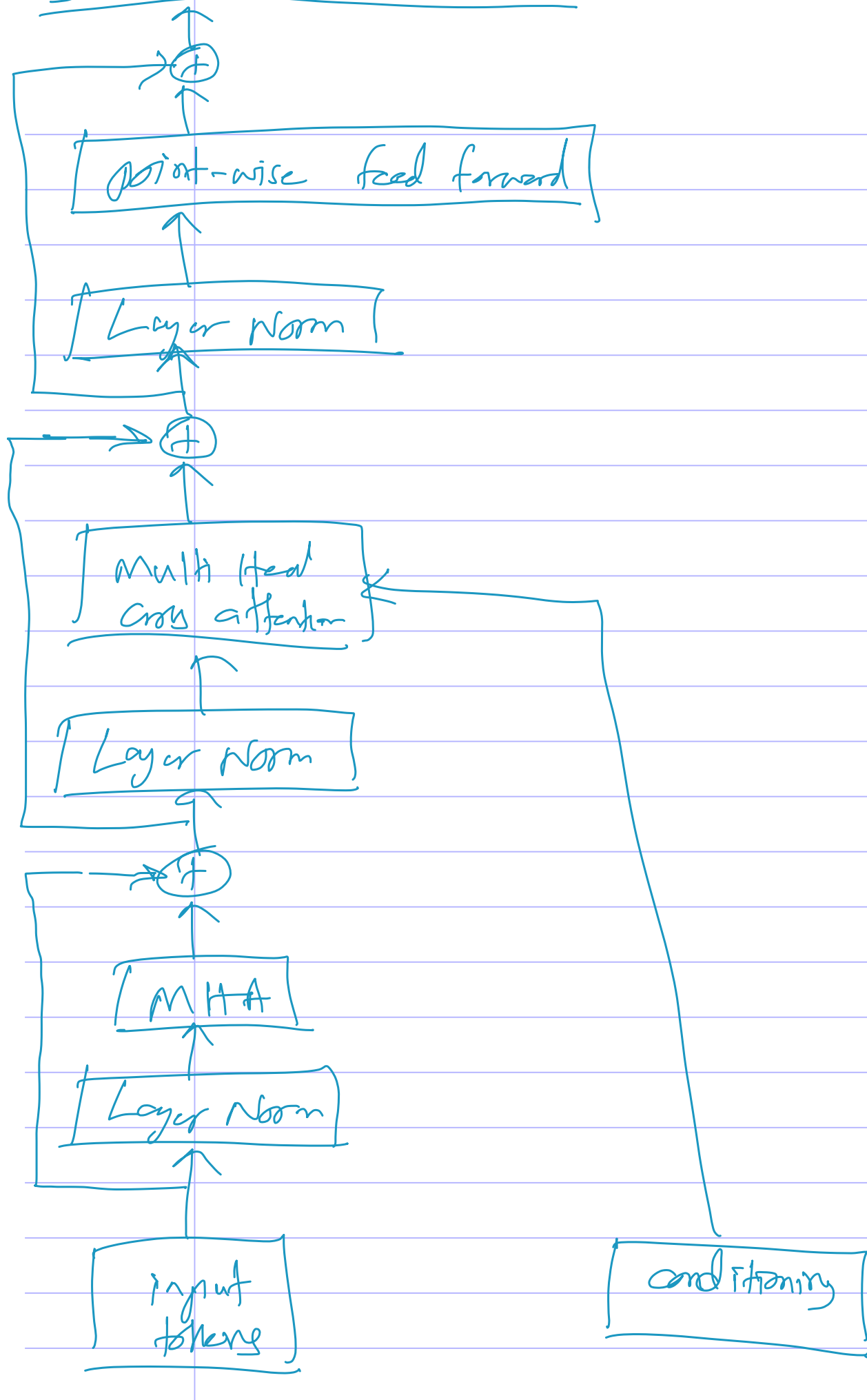


DIT Block Design Choices

① DIT block $\rightarrow$ in-content conditioning



DIT ∇ Cross-attention



DiT Block $\rightarrow$ addLN-zero

